

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Database Management Systems

COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships. (BL1, BL2)*

Activity Tool : Technical Presentation (Low)

Assessment Tool Weightage: 10%

QUESTIONS		
Q. No.	Question	Bloom's Level
1	Create and present an ER diagram for an Online Library System, explaining each component clearly.	BL2 – Understand
2	Prepare a presentation on the EER model, demonstrating generalization and specialization with a University case study.	BL2 – Understand
3	Present the concept of data independence (logical and physical) and its significance in DBMS architecture.	BL2 – Understand
4	Deliver a technical presentation on the software components of a DBMS, explaining query processor, storage manager, and transaction manager.	BL1 – Remember
5	Prepare a presentation comparing strong and weak entities in ER modeling with real-world examples.	BL2 – Understand

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:**RUBRICS FOR CALCULATION:**

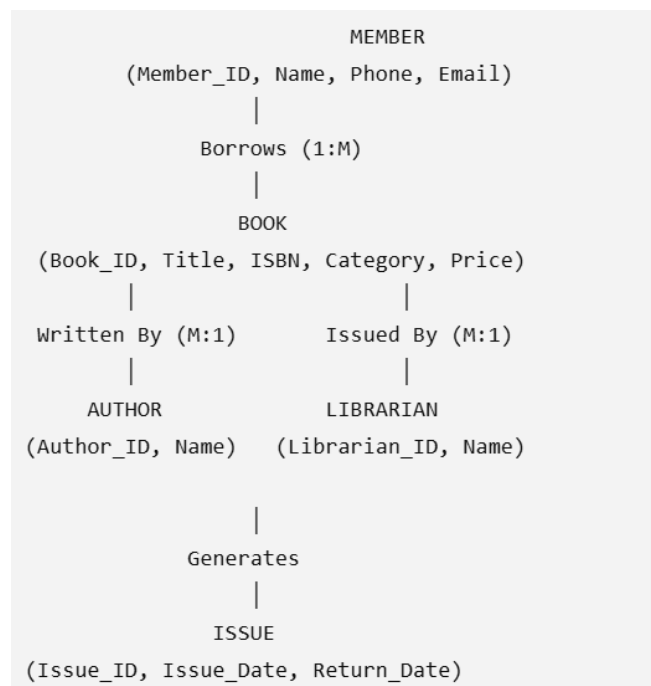
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism ; minimal contribution or ethical awareness

1. Create and present an ER diagram for an Online Library System, explaining each component clearly.

Introduction

An Online Library Management System is used to manage books, members, authors, librarians, and book issue/return details. An Entity-Relationship (ER) Diagram represents the entities, attributes, and relationships in the library database.

ER Diagram



Entities and Attributes

1. Member

Attributes:

- Member_ID (Primary Key)
- Name
- Phone
- Email

2. Book

Attributes:

- Book_ID (Primary Key)
- Title
- ISBN
- Category
- Price

3. Author**Attributes:**

- Author_ID (Primary Key)
- Name

4. Librarian**Attributes:**

- Librarian_ID (Primary Key)
- Name

5. Issue**Attributes:**

- Issue_ID (Primary Key)
- Issue_Date
- Return_Date

Relationships

Relationship	Description	Cardinality
Member – Book	A member borrows books.	1 : M
Book – Author	An author writes books.	M : 1
Librarian – Book	A librarian issues books.	1 : M
Member – Issue	A member has issue records.	1 : M
Book – Issue	A book appears in issue records.	1 : M

Components of the ER Diagram**Entity**

A real-world object in the library system.

- Member
- Book
- Author
- Librarian
- Issue

Attributes

Properties of an entity.

- Member → Name, Phone
- Book → Title, ISBN
- Author → Name

Relationship

Shows how entities are connected.

- Member **borrow**s Book.
- Author **writes** Book.
- Librarian **issues** Book.

Primary Key

A unique identifier for each entity.

- Member_ID
- Book_ID
- Author_ID
- Librarian_ID
- Issue_ID

Conclusion

The ER Diagram for the Online Library System shows how Members, Books, Authors, Librarians, and Issue records are related. It helps organize library data efficiently, reduces redundancy, and simplifies book borrowing and management.

2. Prepare a presentation on the EER Model, demonstrating Generalization and Specialization with a University case study.

Introduction

The Extended Entity-Relationship (EER) Model is an advanced version of the ER model. It includes concepts such as Generalization and Specialization to represent inheritance and complex relationships in a database.

EER Model

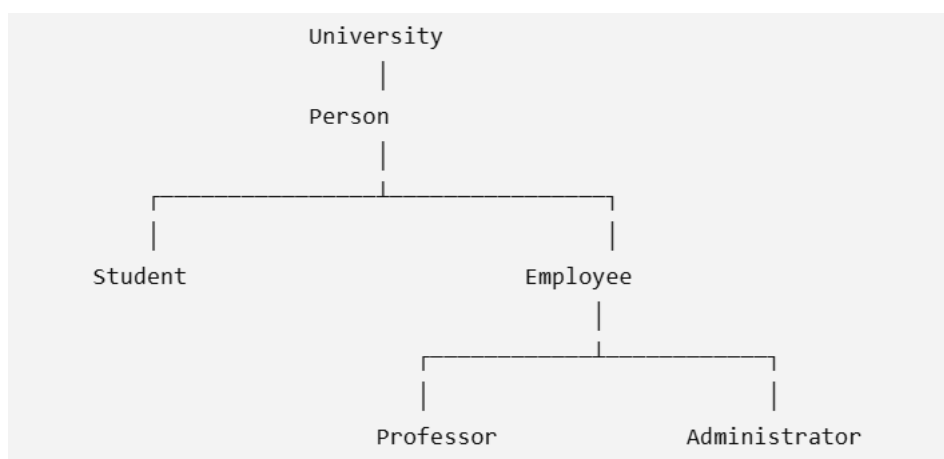
Definition

The EER Model extends the ER model by adding features like Generalization, Specialization, and Inheritance. It helps design complex database systems more effectively.

Features

- Supports inheritance.
- Reduces data redundancy.
- Represents complex relationships.
- Improves database design.

Concept Map



1. Generalization

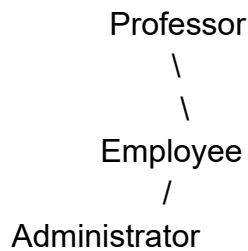
Definition

Generalization is the process of combining similar lower-level entities into a single higher-level entity based on common attributes.

Features

- Bottom-up approach.
- Combines similar entities.
- Creates a common parent entity.

University Example



Both Professor and Administrator have common attributes such as Employee_ID, Name, Address, and Salary, so they are generalized into Employee.

2. Specialization

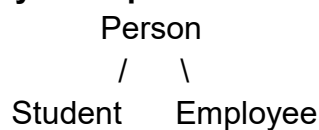
Definition

Specialization is the process of dividing a higher-level entity into lower-level entities based on specific characteristics.

Features

- Top-down approach.
- Child entities inherit parent attributes.
- Adds specialized attributes.

University Example



The Person entity is specialized into Student and Employee.

University Case Study

Super Entity

Person

- Person_ID
- Name
- Address
- Phone

Sub Entities

Student

- Roll_No
- Department
- Course

Employee

- Employee_ID
- Salary

Professor

- Subject
- Qualification

Administrator

- Designation
- Office

Advantages

- Reduces data redundancy.
- Represents inheritance.
- Makes database design simple.
- Improves data organization.

Conclusion

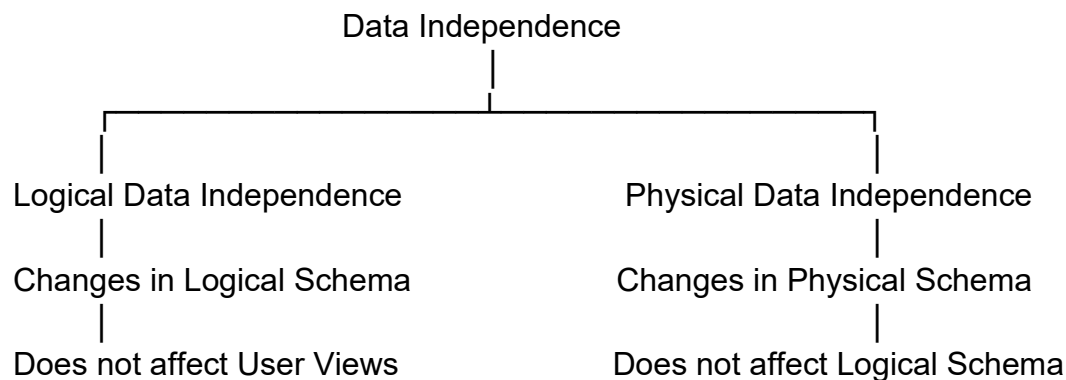
The EER Model enhances the ER model by supporting Generalization and Specialization. In the University case study, the Person entity is specialized into Student and Employee, while Professor and Administrator are generalized under Employee, making the database more organized and efficient.

3 . Present the concept of Data Independence (Logical and Physical) and its significance in DBMS architecture.

Introduction

Data Independence is the ability to modify the database schema at one level without affecting the schema at the next higher level. It is an important feature of the Three-Schema (ANSI/SPARC) Architecture, making databases flexible and easy to maintain.

Concept Map



1. Data Independence

Definition

Data Independence is the ability to change the database schema at one level without affecting the schema at the next higher level.

Types

- Logical Data Independence
- Physical Data Independence

2. Logical Data Independence

Definition

Logical Data Independence is the ability to modify the logical (conceptual) schema without affecting the external (view) schema or application programs.

Example

Adding a new column Email to the Student table without changing existing user applications.

Advantages

- Easy database modification.
- No changes required in application programs.
- Supports database growth.

3. Physical Data Independence

Definition

Physical Data Independence is the ability to modify the physical schema without affecting the logical schema.

Example

Changing data storage from sequential files to indexed files without changing the database tables.

Advantages

- Improves performance.
- Changes storage methods without affecting users.
- Simplifies database maintenance.

Difference Between Logical and Physical Data Independence

Logical Data Independence	Physical Data Independence
Changes the logical schema.	Changes the physical schema.
Does not affect user views.	Does not affect the logical schema.
More difficult to achieve.	Easier to achieve.
Example: Add a new attribute.	Example: Change indexing method.

Significance in DBMS Architecture

- Reduces the impact of database changes.
- Makes database maintenance easier.
- Improves flexibility and scalability.
- Supports efficient database design.
- Protects application programs from schema changes.

Conclusion

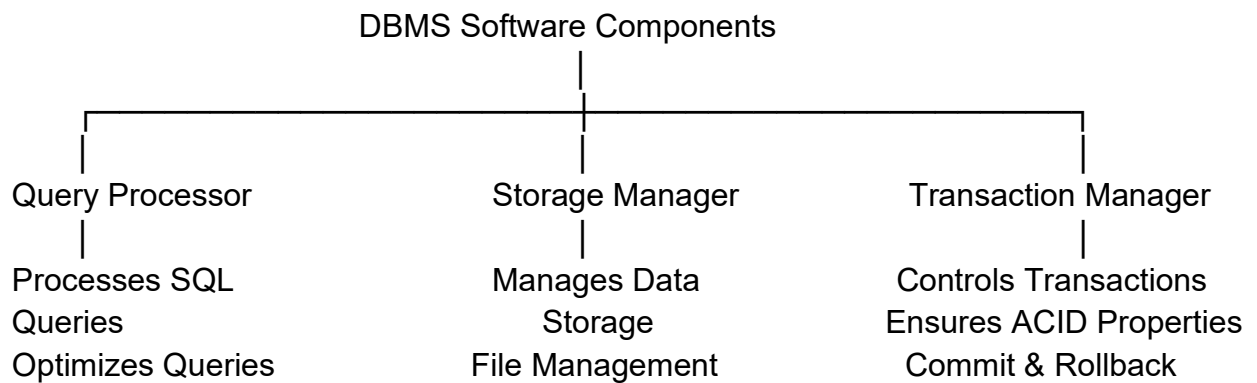
Data Independence is a key feature of DBMS that allows changes to the database without affecting users or applications. Logical Data Independence protects user views from logical changes, while Physical Data Independence protects the logical schema from storage changes, making the database more flexible, reliable, and efficient.

4. Deliver a technical presentation on the software components of a DBMS, explaining Query Processor, Storage Manager, and Transaction Manager.

Introduction

A Database Management System (DBMS) consists of several software components that work together to store, retrieve, update, and manage data efficiently. The three major components are the Query Processor, Storage Manager, and Transaction Manager.

Concept Map



1. Query Processor

Definition

The Query Processor interprets, validates, optimizes, and executes SQL queries submitted by users.

Functions

- Processes SQL queries.
- Checks query syntax.
- Optimizes query execution.
- Retrieves required data.

Example

```
SELECT * FROM Student;
```

The Query Processor processes the SQL statement and returns the required student records.

2. Storage Manager

Definition

The Storage Manager manages the physical storage of data in the database.

Functions

- Stores data on disk.
- Manages files and indexes.
- Retrieves stored data.
- Allocates storage space.

Example

When a new student record is inserted, the Storage Manager stores it in the appropriate database file.

3. Transaction Manager

Definition

The Transaction Manager controls database transactions and ensures that they are executed safely and correctly.

Functions

- Manages transactions.
- Performs Commit and Rollback.
- Maintains data consistency.
- Ensures ACID properties (Atomicity, Consistency, Isolation, Durability).

Example

In a bank transfer, if money is deducted from one account but cannot be credited to another, the Transaction Manager performs a Rollback to restore the original data.

Comparison of DBMS Components

Component	Main Function
Query Processor	Processes and executes SQL queries.
Storage Manager	Manages physical storage of data.
Transaction Manager	Controls transactions and ensures data consistency.

Advantages

- Improves query performance.
- Manages data efficiently.
- Ensures data consistency.
- Supports backup and recovery.
- Provides reliable transaction processing.

Conclusion

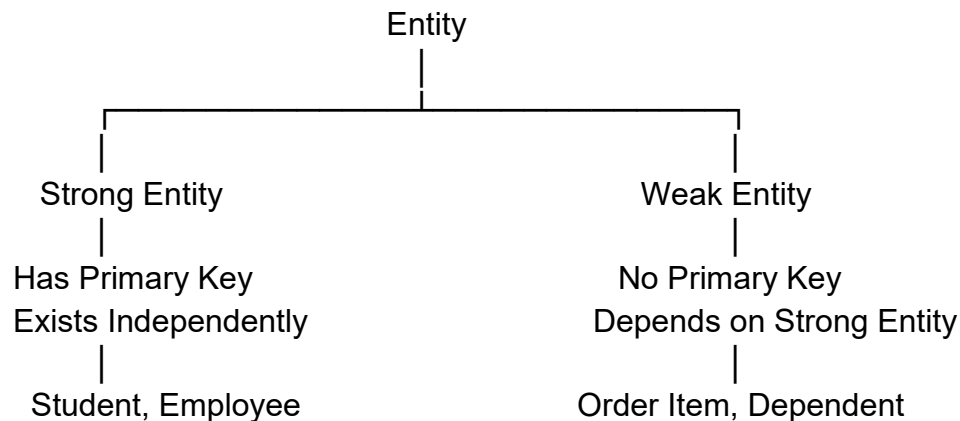
The Query Processor, Storage Manager, and Transaction Manager are the core software components of a DBMS. Together, they ensure efficient query execution, secure data storage, and reliable transaction management, making the database system accurate, secure, and efficient.

5. Create a presentation explaining the concept of Strong and Weak Entities with real-world examples.

Introduction

In an Entity-Relationship (ER) Model, entities are classified into Strong Entities and Weak Entities. A Strong Entity has its own primary key and can exist independently, whereas a Weak Entity depends on a strong entity for its existence.

Concept Map



1. Strong Entity

Definition

A Strong Entity is an entity that has its own Primary Key and can exist independently in the database.

Features

- Has a primary key.
- Exists independently.
- Does not depend on any other entity.
- Represented by a single rectangle in an ER diagram.

Example

Student

Student_ID (PK)	Name	Department
101	Jaya	CSE
102	Priya	ECE

Here, Student is a Strong Entity because it has its own Student_ID.

2. Weak Entity

Definition

A Weak Entity is an entity that cannot exist without a Strong Entity. It does not have a complete primary key and uses the primary key of the strong entity.

Features

- No complete primary key.
- Depends on a strong entity.
- Uses a partial key.
- Represented by a double rectangle in an ER diagram.

Example

Employee → Dependent

Employee_ID (PK)	Employee Name	Dependent_Name	Relationship
E101	Arun	Priya	Daughter
		Ravi	Son

Here, Dependent is a Weak Entity because it cannot exist without the Employee.

Real-World Examples

Strong Entity	Weak Entity
Student	Marksheet
Employee	Dependent
Customer	Order Item
Book	Book Copy
Hospital Patient	Test Report

Difference Between Strong and Weak Entity

Strong Entity	Weak Entity
Has its own primary key.	Has no complete primary key.
Exists independently.	Depends on a strong entity.
Represented by a single rectangle.	Represented by a double rectangle.
Example: Student, Employee	Example: Dependent, Order Item

Advantages

Strong Entity

- Easy to identify.
- Can exist independently.
- Has a unique primary key.

Weak Entity

- Represents dependent data.
- Reduces data duplication.
- Maintains data integrity.

Conclusion

A Strong Entity has its own Primary Key and exists independently, while a Weak Entity depends on a Strong Entity for its existence. Both are essential in database design because they help represent real-world relationships accurately and maintain data consistency.